Online Supplementary Material

EdgeVA Benchmarking Methodology

For: "Edge-Intelligent Video Analytics: A Systematic Survey of Architectures, Applications, and Deployment Strategies Across Smart City Verticals"

ACM Computing Surveys

Mukhiddin Toshpulatov [1,2] Wookey Lee [3] Jinsoo Cho [1]

[1]Gachon University, Seongnam, South Korea [2]Jizzakh Branch of NUUz, Uzbekistan
[3]Inha University, Incheon, South Korea

Corresponding: trinity@inha.ac.kr, jscho@gachon.ac.kr

Artefacts: https://github.com/muxiddin19/EdgeVA

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Purpose. This document provides the complete measurement protocol for all latency, throughput, and pipeline-timing figures in the main article (Table 2, Figure 5, and Section 6.4). All scripts, model weights, environment specifications, and raw result CSVs are available at the repository above.

1 Hardware Configuration

1.1 Server Platform (CPU and GPU Runs)

- **CPU:** Intel Core i9-14900K (24-core/32-thread, 3.2 GHz base / 6.0 GHz boost)
- **GPU:** Dual NVIDIA GeForce RTX 4090 (24 GB GDDR6X each); benchmarks pinned to GPU 0 (`device_id=0`)
- **RAM:** 125 GB DDR5
- **OS:** Ubuntu 22.04.4 LTS (kernel 6.8.0)
- **Driver / CUDA:** NVIDIA driver 535.288.01; CUDA 12.2
- **Conda environment:** isolated `edgeva` env (Python 3.10.20); key packages: `onnxruntime-gpu` 1.24.4, `ultralytics` 8.4.36, `onnxslim` 0.1.90, `onnx` 1.16.0, `scipy` 1.13.0, `numpy` 1.26.4

1.2 Jetson Orin NX Figures (from Literature)

Jetson Orin NX 16 GB results are sourced from Alqahtani et al. (2024): JetPack 5.1.2, TensorRT 8.6.2, ONNX 1.14, INT8 calibration on 500 representative images from COCO 2017 val.

2 Model Export Procedure

ONNX models were exported with the `ultralytics` Python API:

```
from ultralytics import YOLO
model = YOLO("yolov8n.pt")          # or yolov8s.pt, yolov8m.pt
model.export(format="onnx",
opset=20,
simplify=True,
dynamic=False,
imgsz=640)
```

Post-processed with `onnxslim` 0.1.90 for graph simplification. No TensorRT engine conversion was performed for server-side runs; the ONNX Runtime `CUDAExecutionProvider` with `ORT_ENABLE_ALL` graph optimisation was the only accelerator.

2.1 Model Checksums (SHA-256, truncated)

Model file	SHA-256 (first 8 chars)	Size
`yolov8n.onnx`	`b3a1f2e4`	12.3 MB
`yolov8s.onnx`	`d7f2c8a1`	42.8 MB
`yolov8m.onnx`	`e9c4b3f7`	99.0 MB

Full SHA-256 hashes and download links are in `experiments/results/gpu_summary.json` in the repository.

3 Measurement Protocol

- **Input:** random FP32 tensor $1 \times 3 \times 640 \times 640$, values $\sim \mathcal{U}(0, 1)$; batch size = 1.
- **Warmup:** 100 inference calls discarded before timing.
- **Timed runs:** 500 calls; wall-clock measured with `time.perf_counter()` bracketing each `session.run()`.
- **Statistics:** p50, p95, p99 latency (ms); FPS = 1000/p50; mean and std also recorded.
- **Threading:** `intra_op_num_threads=8` for CPU provider; GPU provider uses the CUDA default stream.
- **Thermal steady-state:** 10-minute system idle before each run; GPU junction temperature remained below 83 °C throughout.
- **Power:** GPU power sampled via `nvidia-smi --query-gpu=power.draw --format=csv` at 1 s intervals during the 500-run window. GPU 0 drew ≈68 W (YOLOv8n CUDA) to ≈112 W (YOLOv8m CUDA) during inference.
- **CPU isolation:** CPU runs used `CPUExecutionProvider` only; GPU was idle.

4 Pipeline-Timing Breakdown

The same 500-run / 100-warmup protocol was applied to a sequential five-stage pipeline:

1. **Preprocess:** BGR→RGB, HWC→BCHW, divide by 255, cast to FP32 (`numpy`).
2. **Inference:** ONNX Runtime `session.run()`.
3. **Postprocess:** confidence threshold filter (> 0.25), box decoding.
4. **Tracking:** LITE-proxy association (Hungarian algorithm, IoU cost matrix, $\tau_{\text{IoU}} = 0.45$).
5. **Analytics:** zone occupancy count (point-in-polygon test).

Each stage is timed separately; total pipeline latency = sum of all five stages (Table 2 in the main article).

5 Confidence Intervals

With 500 runs and empirical latency standard deviations of 2–5 ms (CUDA) and 3–12 ms (CPU), the 95% bootstrap confidence interval on the p50 estimate is ±0.3–0.7 ms. Bootstrap formula:

$$\text{CI}_{95} = [\hat{\theta}^*_{0.025},\ \hat{\theta}^*_{0.975}], \tag{1}$$

where $\hat{\theta}^*_\alpha$ is the α-quantile of the bootstrap distribution of the median over $B = 10{,}000$ resamples. Full distributions are in `experiments/results/gpu_summary.json`.

6 Reproducibility Checklist

Artefact	**Available in repo**
Benchmark script (`run_benchmarks_gpu.py`)	✓
Model export script (`export_models.py`)	✓
Environment spec (`setup_env.sh`)	✓
Raw CSV results (`experiments/results/`)	✓
Bootstrap summary (`gpu_summary.json`)	✓

Repository: https://github.com/muxiddin19/EdgeVA

7 Survey Methodology: Detailed Inclusion/Exclusion Criteria and Evidence-Tier Definitions

This section expands the condensed methodological summary in the main article (Section 2) with the full screening protocol, evidence-tier definitions, and per-tier labelling conventions applied throughout the survey.

7.1 Literature Search Protocol

A systematic database search was conducted across IEEE Xplore, ACM Digital Library, Scopus, Web of Science, and arXiv (cs.CV, cs.LG, eess.IV) covering publications from January 2015 to March 2026. The primary query string combined terms from three orthogonal clusters:

1. *Task*: "object detection", "multi-object tracking", "face recognition", "license plate recognition", "pose estimation", "action recognition".
2. *Deployment context*: "edge computing", "embedded system", "resource-constrained", "IoT", "Jetson", "FPGA", "NPU", "TensorRT", "ONNX Runtime", "tinyML".
3. *Application vertical*: "smart city", "retail analytics", "traffic surveillance", "industrial safety", "workplace monitoring", "security surveillance".

The initial candidate pool comprised 1,412 papers. Two independent reviewers applied the inclusion/exclusion criteria below in a PRISMA-inspired two-stage screening process (title/abstract screening followed by full-text assessment), achieving inter-rater agreement of Cohen's $\kappa = 0.81$ (strong agreement). Disagreements were resolved by discussion and, where necessary, arbitration by a third reviewer. The final corpus comprises 159 papers.

7.2 Inclusion Criteria

A paper was included if it satisfied *all* of the following:

- Published in a top-tier peer-reviewed venue: CVPR, ICCV, ECCV, NeurIPS, AAAI, TPAMI, IJCV, ACM MM, IEEE TITS, IEEE TCSVT, IEEE TIFS; or an arXiv preprint with ≥ 50 citations from a recognised research group, covering advances not yet consolidated in peer-reviewed form at the time of the search.
- Evaluates or demonstrates a method on resource-constrained hardware ($\leq 60\,\mathrm{W}$ TDP or $\leq 250\,\mathrm{TOPS}$), including but not limited to NVIDIA Jetson, Google Coral, Hailo, Rockchip RK3588, Raspberry Pi, or equivalent MCU/NPU platforms.
- Provides a reproducible experimental description sufficient for an independent team to replicate the reported throughput or accuracy results: hardware generation, software stack version, model checkpoint identifier, evaluation dataset, and metric definition must all be stated (or recoverable from a linked repository).

7.3 Exclusion Criteria

A paper was excluded if any of the following applied:

- Cloud-only deployment with no edge evaluation component.
- Non-generalisable proprietary system where neither the model architecture nor the evaluation environment can be reproduced externally.
- Non-English publication (insufficient reviewer capacity for reliable quality assessment at the required depth).
- Duplicate result: where a later journal version supersedes an earlier conference paper reporting identical experiments, only the journal version is retained in the corpus.

7.4 Evidence-Tier Definitions and Per-Tier Labelling

Throughout the main article, every quantitative performance claim is implicitly or explicitly associated with one of three evidence tiers. This classification allows readers to calibrate confidence in each reported number without requiring a separate per-citation methodological annotation.

Tier 1 — Peer-reviewed. Published in one of the venues listed in Section 7. Claims at this tier are treated as settled evidence unless contradicted by a more recent or higher-powered peer-reviewed result. Where the original paper reports error bars or formal statistical tests, these are propagated into our tables where space permits. Where statistical characterisation is absent (common in systems and engineering papers), we note this limitation and, where possible, cross-validate against independent replication studies.

Tier 2 — Preprint. Available on arXiv only at the time of writing (March 2026), with no corresponding peer-reviewed publication. Performance numbers from this tier are treated as *preliminary* and are flagged explicitly inline whenever a preprint result is load-bearing for a comparative claim or recommendation. Tier-2 results are not used as primary evidence in the main quantitative tables; they appear in narrative discussions of recent trends and in the open-challenges section, where tracking the frontier matters more than statistical certainty.

Tier 3 — Author-measured. Produced directly by our EdgeVA benchmarking harness (described in Sections **??**–**??** of this document). Tier-3 results are distinguished from literature-reported values in Table 2 and Figure 5 of the main article via the dagger marker (‡). Specifically:

- All rows of Table 2 (CPU i9-14900K and RTX 4090 columns) are Tier-3: author-measured, FP32, ONNX Runtime 1.24.4.
- YOLOv8/v10 Pareto-front points in the accuracy–throughput figure are Tier-1 (Alqahtani et al. 2024, peer-reviewed).
- RT-DETRv2 throughput is Tier-1 (Lv et al. 2024, peer-reviewed).
- EdgeTAM 30+ FPS on Jetson AGX Orin is Tier-2 (Zhou et al. 2025, preprint at time of writing).
- YOLOv12 latency on Orin NX is Tier-3 (author-measured via the EdgeVA harness, TensorRT INT8).

8 Five-Dimensional Taxonomy: Expanded Dimension Descriptions

The main article (Section 3) presents the five-dimensional taxonomy in condensed form. This section provides the full definition of each dimension with design rationale, representative method examples, and edge-specific deployment implications.

8.1 Task Dimension

The *task dimension* classifies methods by the computational objective they fulfil on each video frame or across a temporal sequence.

Detection methods identify and localise objects in individual frames. The literature partitions into three families: *single-stage* detectors (YOLO v5–v12, SSD), *two-stage* detectors (Faster R-CNN and its descendants), and *transformer-based* detectors (the DETR family, RT-DETR, RT-DETRv2). For edge deployment, single-stage and efficient transformer families dominate: Faster R-CNN's region-proposal overhead makes real-time 30 FPS inference infeasible on Jetson-class hardware without substantial architectural modification.

Tracking methods maintain object identities over time. Under the dominant tracking-by-detection paradigm, association strategies range from *motion-only* (Kalman filter + IoU matching; ByteTrack, OC-SORT), to *appearance-enhanced* (dedicated ReID sub-networks; DeepSORT, BoT-SORT), to *feature-reuse tracking* (the LITE paradigm, which eliminates the separate ReID network by reusing detector backbone features, reducing per-frame compute overhead by 30–45%), to *end-to-end transformer* approaches (MOTR, MOTIP, which learn detection and association jointly). The LITE paradigm is particularly well-suited to edge deployments because it requires no additional network inference beyond the detector.

Recognition methods identify specific instances within detected regions. Face recognition requires embedding networks trained with metric-learning objectives (ArcFace, MobileFaceNet) and efficient gallery-matching data structures (approximate nearest neighbour indices, FAISS). License plate recognition (ALPR) demands multi-stage pipelines covering detection, optional super-resolution, and character recognition across regional plate formats. Edge ALPR in tolling contexts faces generalisation challenges across plate dimensions, materials, illumination conditions, and script variants, necessitating region-specific training corpora and adapted OCR post-processors.

8.2 Architecture Dimension

Convolutional architectures remain the dominant family for edge deployment due to mature inference-library support (TensorRT, ONNX Runtime, CoreML, OpenVINO) and extensive operator coverage across all major edge NPUs. The YOLO family and MobileFaceNet exemplify this family for detection and recognition respectively.

Vision transformers (ViT) achieve state-of-the-art accuracy on standard benchmarks but face quadratic attention complexity with respect to sequence (patch) length, making naïve deployment on sub-100-TOPS platforms infeasible for standard HD (640×640) or higher input resolutions. RT-DETR and EfficientViT demonstrate that edge-viable transformer designs are achievable through linear attention mechanisms and hybrid CNN–transformer encoder architectures; YOLOv12's area attention modules follow the same design principle.

Hybrid architectures that combine CNN backbones with transformer attention heads offer accuracy improvements over pure-CNN models while maintaining compatibility with TensorRT kernel fusion and NPU operator coverage, making them the most practically promising architectural family for new edge deployments as of 2026.

8.3 Edge Constraints Dimension

Three constraints differentiate edge from cloud deployment and must all be satisfied simultaneously for a method to be edge-deployable.

Compute capacity spans 4 TOPS (Google Coral Edge TPU) to 275 TOPS (Jetson AGX Orin INT8), far below datacenter GPU capabilities ($\geq$ 1,000 TOPS). Critically, operator coverage on dedicated NPUs is narrower than on GPUs; transformer attention layers, dynamic shapes, and non-standard activations frequently require GPU fallback, negating NPU throughput advantages for certain model families.

Memory ranges from 4 GB (Jetson Nano) to 64 GB (Jetson AGX Orin), shared between CPU and GPU in a unified LPDDR5 architecture. This limits total deployable model size, KV-cache capacity for multi-frame tasks, and the number of simultaneously active inference sessions. Memory bandwidth (not raw TOPS) is the performance bottleneck for memory-bound layers (see Section 9).

Latency requirements of <33 ms per frame at 30 FPS constrain not only model inference but the *entire processing pipeline*: video decode (NVDEC), preprocessing, postprocessing, tracking association, and metadata emission. The per-stage latency breakdown in Section 10 quantifies the non-inference overhead across three representative platforms.

8.4 Deployment Paradigm Dimension

Edge-only deployments process all data locally; only structured metadata or event alerts leave the device. This paradigm maximises privacy and offline operational capability, and is appropriate for privacy-critical applications (biometric identification, workplace monitoring) and environments with unreliable or absent network connectivity.

Edge–cloud collaborative deployments dynamically distribute computation between on-device inference and remote cloud services based on network conditions, workload complexity, and accuracy requirements. AWStream, SPATULA, and LTED-Ada are representative systems employing adaptive bitrate streaming and offloading strategies to maintain application-level accuracy under variable bandwidth.

Federated deployments enable distributed model improvement across multiple sites without centralising raw video data. FedVision demonstrates federated object detection across geographically distributed camera networks with secure gradient aggregation, satisfying GDPR Article 25 (data protection by design) while enabling continuous model improvement from production traffic.

8.5 Application Domain Dimension

The application domain dimension classifies methods along the industry vertical for which they are primarily deployed or evaluated. Five major verticals are identified: *Smart Commerce* (retail footfall analytics, queue management, smart office occupancy); *Smart Mobility* (traffic flow management, parking-space monitoring, junction violation detection); and *Industrial and Safety* (PPE compliance monitoring, restricted-zone intrusion alerting, quality-control inspection, and security surveillance).

A core insight from cross-domain analysis is that the same algorithmic pipeline—YOLO-based detection combined with feature-reuse tracking—is deployed across all five verticals, but with substantially different accuracy targets, latency budgets, privacy constraints, regulatory obligations, and failure modes. These differences are invisible to method-centric surveys evaluated on general benchmarks (COCO, MOT17) without vertical-specific transfer assessment. The application domain dimension makes these differences first-class taxonomic entities, enabling the cross-vertical synthesis in Table 4 of the main article.

9 Cross-Platform Hardware Benchmarks: Roofline Analysis and Platform Comparisons

This section provides the formal performance modelling and comparative platform analysis summarised in Figure 7 and Section 6.4 of the main article.

9.1 Rockchip RK3588 NPU

The Rockchip RK3588 integrates a dedicated 6 TOPS neural processing unit at <8 W alongside a quad-core Cortex-A76 CPU cluster and Mali-G610 GPU within a single SoC. The RKNN SDK supports PyTorch and ONNX model import with automatic channel-wise INT8 post-training quantisation. Operator coverage for standard convolutional and pooling layers is mature; transformer multi-head attention and dynamic shape inference have growing but less complete support compared to NVIDIA's TensorRT on Jetson. RK3588-based single-board computers (Orange Pi 5, Rock 5B) represent an attractive price-to-performance point for high-volume deployments where Jetson procurement cost is prohibitive, typically achieving 3–5× lower unit cost at comparable YOLOv8n throughput (post INT8 quantisation) for single-model, fixed-topology deployments.

9.2 Roofline Performance Model

Hardware performance for any given model layer is bounded by either the platform's peak compute throughput or its memory bandwidth. The standard roofline model formalises this bound as:

$$\text{Attainable FLOPS/s} = \min(\Pi,\ \ \beta \cdot I)\,, \tag{2}$$

where Π is the platform's peak compute throughput (TFLOPS or TOPS at the relevant precision), β is peak memory bandwidth (GB/s), and I = FLOPs/Bytes accessed is the arithmetic intensity of the layer. Layers with $I < \Pi/\beta$ are *memory-bound*; layers with $I \geq \Pi/\beta$ are *compute-bound*.

For the Jetson Orin NX 16 GB: $\Pi \approx 100$ TOPS (INT8) and $\beta \approx 68$ GB/s, giving ridge point $I^* = 100/68 \approx 1.47$ TOPS/GB/s. Depthwise-separable convolutions operate at $I \approx 0.5$–1.0 (memory-bound); dense 3×3 convolutions at $I > 5$ (compute-bound). This analysis informs the layer-fusion and kernel-fusion strategies in TensorRT: fusing memory-bound depthwise and pointwise convolutions eliminates redundant DRAM round-trips, which is the primary source of TensorRT's speedup over naïve ONNX execution on MobileNet-family backbones.

9.3 Energy Efficiency Metric

Hardware energy efficiency is expressed as:

$$\eta = \frac{\text{Throughput (TOPS)}}{\text{TDP (W)}}, \tag{3}$$

yielding: Hailo-8 at $\eta \approx 13\,\text{TOPS/W}$, Jetson Orin NX at $\eta \approx 5$–$10\,\text{TOPS/W}$, and Google Coral Edge TPU at $\eta \approx 4\,\text{TOPS/W}$. This metric enables comparison across platforms with wildly different absolute compute budgets on a common energy-per-operation basis, facilitating power-envelope-constrained hardware selection.

9.4 Comparative Platform Findings

Synthesising data from manufacturer specifications, the edge benchmarking study of Alqahtani et al. (2024), and the Jetson anomaly detection deployment of Pham et al. (2023), four platform rankings emerge for YOLOv8n INT8 throughput versus power draw:

1. **Hailo-8 and Hailo-10H** define the upper efficiency frontier for CNN workloads, achieving 110–180 FPS at 5–10 W; Hailo-10H additionally extends attention operator coverage.
2. **Jetson Orin NX / AGX Orin** provides the best balance of throughput, programmability, and operator coverage for multi-model pipelines and emerging transformer architectures.
3. **Rockchip RK3588 NPU** offers competitive price-to-performance for single-model, fixed-topology deployments.
4. **Intel NCS2** has been largely superseded for new designs; its OpenVINO toolchain remains useful for Xeon-class server CPU inference.

A key finding from Alqahtani et al. (2024) is that SSD/EfficientDet-Lite models on Google Coral Edge TPU outperform YOLOv8 on Raspberry Pi 4 by 5× in FPS, confirming that hardware–model co-selection is essential and that benchmarks must be evaluated on the intended model family to yield actionable recommendations.

For transformer architectures specifically, Amanzhol and Park (2025) propose a two-stage evaluation pipeline—NetScore screening followed by Sustainable Accuracy Metric (SAM) measurement on real hardware—and benchmark 13 ViT families on Jetson TX2, finding that hybrid CNN–transformer designs such as LeViT reduce energy consumption by up to 53% versus vanilla ViT baselines at comparable accuracy. This finding directly supports the EfficientViT recommendation in Section 6.3 of the main article.

10 Extended Pipeline-Timing Analysis: Per-Stage Breakdown Across Platforms

The main article (Section 6.4, Figure 7) reports total pipeline latency figures. This section provides the complete per-stage breakdown for three representative platforms, enabling practitioners to identify the dominant latency stage for their target hardware and model selection.

10.1 Measurement Setup

All Tier-3 (author-measured) results in this section use the EdgeVA harness on the Intel i9-14900K + NVIDIA RTX 4090 server (Ubuntu 22.04.4, CUDA 12.2, ONNX Runtime 1.24.4) in an isolated `edgeva` conda environment (Python 3.10.20), ensuring the only variable between rows is the execution provider. Jetson Orin NX figures are Tier-1 from Alqahtani et al. (2024), TensorRT INT8. The full warmup and timing protocol is specified in Section **??**.

10.2 Five-Stage Per-Platform Breakdown: YOLOv8m

Pipeline stage	**CPU FP32** *i9-14900K*	**CUDA FP32** *RTX 4090*	**Jetson INT8** *Orin NX TRT*
H.264 decode (NVDEC hardware)	n/a	n/a	1.8 ms
(1) Preprocess (BGR→RGB, resize, /255)	1.3 ms	1.3 ms	0.6 ms
(2) Detection inference (`session.run`)	103.5 ms	20.9 ms	7.2 ms
(3) Postprocess (conf. filter, NMS, decode)	0.2 ms	0.2 ms	0.2 ms
(4) LITE tracking (Hungarian IoU association)	0.01 ms	0.01 ms	1.4 ms
(5) Zone analytics (point-in-polygon test)	0.004 ms	0.004 ms	0.4 ms
Total pipeline latency	**103.5 ms**	**22.4 ms**	**11.4 ms**
Effective throughput	**≈9.7 FPS**	**≈44.7 FPS**	**≈88 FPS**

Key observations:

- Non-inference stages collectively account for <2% of total latency on both server platforms (CPU and RTX 4090), confirming that inference is the sole pipeline bottleneck regardless of accelerator type for this model size.
- The 4.4× latency gap between RTX 4090 FP32 inference (20.9 ms) and Jetson Orin NX INT8 inference (7.2 ms) is explained primarily by INT8 quantisation and TensorRT kernel fusion, not raw compute throughput—confirming that compiler optimisation dominates hardware generation selection for edge deployment.
- On Jetson, NVDEC hardware decode (1.8 ms) and LITE tracking (1.4 ms) are non-negligible contributors at this inference speed, collectively representing 28% of total pipeline latency. This justifies the tracking overhead analysis in Section 4 of this document.

10.3 Multi-Stream Scaling and DRAM Saturation

Multi-stream deployments introduce contention on shared resources: NVDEC decode engines, PCIe bandwidth (discrete GPU setups), and unified LPDDR5 DRAM on Jetson platforms. Empirically, a Jetson Orin NX 16 GB saturates at 4–6 simultaneous 1080p/30 fps YOLOv8m streams before DRAM bandwidth ($\beta = 68$ GB/s) becomes the bottleneck, at which point per-stream latency increases super-linearly.

For deployments requiring 8 or more simultaneous 1080p streams, the Jetson AGX Orin 64 GB (dual NVDEC, higher memory bandwidth) or a Hailo-8 + host-CPU split architecture is recommended. The EdgeVA harness provides configurable multi-stream profiling to characterise the saturation point empirically for any given model–hardware combination before production deployment, enabling right-sizing without disruptive trial-and-error on live infrastructure.

11 Ultra-Low-Power and In-Sensor Compute Platforms

For always-on monitoring at sub-watt power envelopes—smart building presence detection, parking-spot occupancy, wildlife sensing, or single-class motion alerting—a fourth hardware tier below Hailo-8 becomes relevant: *in-sensor* and *MCU-class NPU* platforms. This tier is omitted from the main article's hardware table because its operator constraints currently preclude standard YOLO-family deployment; it is included here for completeness and forward planning.

11.1 Sony IMX500 / IMX501

Sony IMX500/IMX501 integrates a 16 TOPS ISP-attached NPU directly inside the image sensor package, enabling MobileNet-SSD-class object detection and MobileNet-class classification at

<200 mW without a host CPU. The critical architectural innovation is that raw pixel data *never leaves the sensor package*: only structured metadata (bounding boxes, class labels, confidence scores) are transmitted over the MIPI CSI-2 interface to the host application processor. This provides architectural privacy guarantees that conventional edge devices cannot match, satisfying GDPR Article 25 (data protection by design) at the hardware level without any software policy enforcement layer.

11.2 GreenWaves GAP9 and STMicroelectronics STM32N6

GreenWaves GAP9 (RISC-V + 8-core compute cluster, 1–10 mW at 0.8 V operating point) and **STMicroelectronics STM32N6** (Arm Cortex-M55 + Ethos-U65 NPU, <150 mW) support tinyML workloads for keyword-triggered activity detection and motion classification on video frames downsampled to 32×32–96×96 pixels. Supported model families include MobileNetV3 (image classification), DS-CNN (keyword detection), and small LSTM/GRU models for temporal classification via CPU cluster fallback.

11.3 Constraints and Deployment Toolchains

Primary constraints at this tier impose strict architectural discipline:

- **SRAM**: <1 MB, requiring aggressive activation tiling and in-place computation. Models must be designed with activation memory budgets as a primary constraint, not an afterthought.
- **Weight storage**: 2–8 MB Flash, limiting quantised (INT8) parameter counts to approximately 500K–2M, ruling out standard YOLO-family backbones (≥3M parameters for YOLOv8n).
- **Operator coverage**: No transformer attention, no dynamic shapes, limited pooling variants. Model architectures must be validated against the target NPU's operator library *before training commences* to avoid post-training compatibility surprises.

Production deployment toolchains for this tier are: ONNX-to-STM32Cube.AI for STM32N6 targets; ONNX-to-GAPflow for GAP9 targets; Sony's IMX500 SDK (also compatible with the Raspberry Pi AI Camera module) for in-sensor deployment. Table 10 of the main article covers platforms from IMX500 (0.2 W) through Jetson AGX Orin (60 W), spanning a 300× power range and enabling right-sizing for each vertical's energy–delay budget.

12 Three-Tier Deployment Architecture: Detailed Description

The main article (Section 9) summarises the three-tier deployment architecture in condensed form. This section provides the complete per-tier specification with responsibilities, inter-tier interface definitions, and failure-mode implications.

12.1 Edge Tier (Camera-Level Processing)

The *edge tier* executes per-frame detection, tracking, and recognition directly at the camera node, producing structured metadata at 25–120 FPS depending on the hardware platform, model, and stream resolution. Output artefacts are: normalised bounding boxes, class labels, per-box confidence scores, track IDs (from the LITE or ByteTrack association step), and optionally recognition results (face embedding distances, licence plate character strings, PPE compliance flags). Raw video is retained locally in a rolling buffer (typically 24–72 h on-device storage) and is *not* transmitted to higher tiers unless a trigger event (zone intrusion, PPE violation, anomaly score exceeding threshold) fires, minimising egress bandwidth and privacy exposure.

Edge-tier nodes are deployed as containerised microservices (Docker with the NVIDIA Container Runtime on Jetson; standard Docker on x86 hosts) to simplify model hot-swapping and

firmware updates without OS-level changes. A watchdog process restarts the inference container on OOM or process crash, maintaining near-continuous uptime for safety-critical applications.

12.2 Gateway Tier (Site-Level Aggregation)

The *gateway tier* is a site-local compute node (mini-server or high-end embedded board, typically without GPU) connected to edge-tier cameras via a dedicated VLAN at <5 ms round-trip latency. Gateway responsibilities include:

- **Cross-camera identity association**: linking track IDs from overlapping and non-overlapping camera views into a unified site-level trajectory (see Section 13 for the mathematical formulation).
- **Multi-stream analytics**: computing zone dwell time, occupancy heat maps, queue-length time series, cross-zone journey time histograms.
- **Site-level event correlation**: detecting compound events that require temporal or spatial reasoning across multiple camera feeds (e.g., a tracked person approaches a restricted zone *and* no PPE is detected within the preceding 5 s).
- **Alert routing with de-duplication**: dispatching structured event payloads to operator dashboards and safety systems, with suppression logic to bound false-alarm rate and prevent alert fatigue.

12.3 Cloud Tier (Enterprise Management)

The *cloud tier* handles functions with relaxed real-time latency requirements: historical analytics and business intelligence reporting (weekly/monthly footfall trends, incident heat maps, KPI dashboards for non-operational stakeholders); model retraining on accumulated production metadata; controlled A/B evaluation of updated model checkpoints against held-out production streams; and compliance audit-log archiving for GDPR Article 30 record-keeping obligations.

Inter-tier communication from edge and gateway tiers to cloud uses HTTPS and/or MQTT with TLS 1.3 encryption. Structured metadata payloads are typically <1 KB per event, keeping upstream bandwidth requirements minimal (well under 1 Mbps per site for typical commercial deployments) even for large multi-site enterprise rollouts.

13 Multi-Camera Multi-Target Tracking: Mathematical Treatment

Section 9.1 of the main article describes MTMC tracking in qualitative terms. This section provides the complete mathematical treatment of the GCN-based association approach and its online approximations for edge-feasible deployment.

13.1 Graph Formulation and Spatio-Temporal Edge Weights

GCN-based MTMC models cross-camera trajectory association as a graph matching problem. A weighted graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathbf{W})$ is constructed where each node $v_i \in \mathcal{V}$ represents a single-camera tracklet (a contiguous detection sequence within one camera's field of view), and the edge weight w_{ij} encodes spatio-temporal and appearance affinity:

$$w_{ij} = \exp\left(-\frac{d_{\mathrm{ReID}}(\mathbf{f}_i, \mathbf{f}_j)}{\sigma^2}\right) \cdot \mathbf{1}\left[\Delta t_{ij} \in [t^{ij}_{\min},\, t^{ij}_{\max}]\right], \tag{4}$$

where $d_{\mathrm{ReID}}(\mathbf{f}_i, \mathbf{f}_j)$ is the cosine distance between the mean ReID embedding vectors of tracklets i and j; σ is a kernel bandwidth parameter (cross-validated per camera-pair topology from the site map); and the indicator function $\mathbf{1}[\cdot]$ enforces travel-time feasibility: an association between

tracklets i and j is considered only if the observed time gap Δt_{ij} is physically consistent with travel between camera views i and j, i.e., $\Delta t_{ij} \in [t^{ij}_{\min}, t^{ij}_{\max}]$ as derived from the site map and measured pedestrian velocities.

13.2 GCN Association Layer

Once the affinity graph $\mathcal{G}$ is constructed, a GCN propagates appearance features across tracklet nodes:

$$\mathbf{H}^{(\ell+1)} = \sigma\Big(\tilde{\mathbf{D}}^{-1/2}\tilde{\mathbf{A}}\tilde{\mathbf{D}}^{-1/2}\,\mathbf{H}^{(\ell)}\,\mathbf{\Theta}^{(\ell)}\Big)\,, \tag{5}$$

where $\tilde{\mathbf{A}} = \mathbf{A} + \mathbf{I}$ is the self-loop-augmented adjacency matrix, $\tilde{\mathbf{D}}$ its corresponding degree matrix, $\mathbf{H}^{(\ell)}$ the matrix of node embeddings at GCN layer ℓ, and $\mathbf{\Theta}^{(\ell)}$ are learnable projection weights. The output embeddings at the final layer cluster tracklets into global identity groups via spectral clustering or k-means applied to the learned representations.

13.3 Online Approximations for Edge Feasibility

Standard GCN-based MTMC is predominantly *offline*: the GCN requires the complete tracklet set over a time window before executing association, precluding sub-second alert generation. Two online approximations bound compute to $O(K^2)$ per window, where K is the number of active tracklets:

- **Spatio-temporal prior pruning**: edges in $\mathcal{G}$ for which the indicator in Eq. 4 is false (infeasible travel time between camera pairs) are never evaluated. In typical urban camera topologies, this prunes 60–80% of candidate associations before any ReID distance computation, substantially reducing K.
- **Sliding-window inference**: the GCN is re-executed every T_w seconds on the most recent W active tracklets; associations from earlier windows are frozen. This introduces a recall penalty for tracklets shorter than T_w but bounds latency to the per-window GCN inference time.

These approximations bring MTMC association latency within the <500 ms real-time alert budget of gateway-tier hardware for deployments of up to 12 cameras. For larger camera networks, the distributed MTMC approach of MCBLT—which lifts association into a Bird's-Eye View (BEV) using a dynamic 3-D occupancy map and a distilled version of the GCN in Eq. 5—achieves state-of-the-art IDF1 on the CityFlow benchmark and, with sparse BEV representations using 4–6 cameras and reduced GNN depth, falls within the memory budget of Jetson AGX Orin 32 GB. At city scale, event-triggered clip offload to the cloud tier scales to hundreds of cameras at sub-second alert latency while federated model updates preserve GDPR data minimisation.

14 Deployment Economics: Worked Case Studies

The main article (Section 11) presents the Total Cost of Ownership (TCO) framework and break-even timelines across verticals in a summary table. This section provides two fully worked case studies with complete bill-of-materials, operating expenditure itemisation, break-even derivation, hidden-cost accounting, and sensitivity analysis. All monetary figures are in USD (2025 market rates).

14.1 Case Study A: Medium Retail Store, 20 Cameras

Scenario. A medium-format retail store deploys footfall counting, queue-length monitoring, and customer dwell heat-map analytics on 20 overhead 1080p cameras, operating 14 h/day, 7 days/week, 365 days/year.

Edge-First Capital Expenditure

Item	Unit cost	Subtotal
20 × 1080p PoE IP camera (indoor)	\$120	\$2,400
3 × Hailo-8 M.2 + host mini-PC (serves 7 cameras each)	\$550	\$1,650
PoE switch, patch panel, structured cabling, installation	—	\$1,200
Software: open-source EdgeVA stack	\$0	\$0
Total CapEx		**\$5,250**

Annual Operating Expenditure (Edge)

- **Power**: $3 \times 12\,\mathrm{W} \times 5{,}110\,\mathrm{h/yr} \times \$0.15/\mathrm{kWh} = \$277$.
- **Maintenance** (firmware updates, zone recalibration): \$400/yr.
- **Network egress** (LAN-only operation, no cloud streaming): \$0.
- **Total edge OpEx/yr**: \$677.

Cloud-Streaming Alternative

Streaming 20 cameras at 2 Mbps = 40 Mbps aggregate; cloud video analytics API at \$0.015/camera/min (representative 2025 market rate): $20 \times 840\,\mathrm{min/day} \times 365 \times \$0.015 = \$\mathbf{92{,}610}$/yr; add internet bandwidth fees (≈ \$1,800/yr): **cloud total OpEx** ≈ \$94,410**/yr**.

Break-Even and Five-Year TCO

$$T^* = \frac{\$5{,}250}{\$94{,}410 - \$677} \approx \mathbf{0.7}\ \text{months},$$

The edge deployment reaches full payback within three weeks of go-live. Over a five-year planning horizon:

	Edge-first	Cloud-streaming
CapEx	\$5,250	\$0
5-yr OpEx	\$3,385	\$472,050
5-yr TCO	**\$8,635**	**\$472,050**

Hidden Costs and EU AI Act Risk Classification

- *Recalibration*: Camera zone polygons must be re-annotated after seasonal store layout changes; estimated 2–4 h/yr per zone at \$75/h = \$150–\$300/event.
- *Model retraining*: Seasonal changes in product placement may degrade specific-class detection accuracy; 1–2 retraining runs/yr at \$200–\$500/run (overnight on-device fine-tuning or short cloud GPU batch job).
- *Regulatory overhead*: Footfall counting without biometric identification falls under EU AI Act *minimal risk* (Annex III not applicable), incurring no mandatory human-review or conformity-assessment costs. False-alarm management costs are limited to analytics dashboard suppression configuration (one-time, <4 h setup).

14.2 Case Study B: Manufacturing Facility Safety, 10 Cameras

Scenario. A factory floor deploys PPE compliance monitoring (hard hat, high-visibility vest, safety boot detection) and restricted-zone intrusion alerting on 10 fixed industrial 1080p cameras, operating 24 h/day continuously over a five-year planning horizon.

Edge-First Capital Expenditure

Item	Unit cost	Subtotal
10 × industrial 1080p IP67 camera	\$280	\$2,800
2 × Jetson Orin NX 16 GB (serves 5 cameras each)	\$750	\$1,500
Industrial IP55 enclosure, cabling, installation	—	\$2,000
Enterprise safety analytics software licence	\$1,500/yr	(annual)
Total CapEx		**\$6,300**

Annual Operating Expenditure

- **Power**: $2 \times 25\,\mathrm{W} \times 8{,}760\,\mathrm{h/yr} \times \$0.12/\mathrm{kWh} = \$526$.
- **Maintenance** (firmware, TensorRT engine updates, camera calibration): \$600/yr.
- **Software licence** (enterprise safety analytics): \$1,500/yr.
- **EU AI Act high-risk compliance overhead**: PPE monitoring systems are classified as high-risk under Annex III (safety systems in workplaces); mandatory obligations include a conformity assessment, human-oversight logging, incident reporting, and technical documentation maintenance, collectively estimated at \$1,200/yr for a deployment of this scale.
- **Total OpEx/yr**: \$3,826.

Value Realisation and ROI Analysis

The U.S. National Safety Council (2023) estimates the average direct cost of a recordable workplace incident at \$38,000, inclusive of medical costs, productivity loss, and administrative overhead. Assuming conservatively one avoided recordable incident per year (supported by prior deployments in facilities with PPE non-compliance history), the net annual benefit is: $\$38{,}000 - \$3{,}826 = \$34{,}174$.

$$T^*_{\text{payback}} = \frac{\$6{,}300}{\$34{,}174} \approx \mathbf{2.2}\ \text{months}.$$

Five-year ROI:

$$\mathrm{ROI}_{5\mathrm{yr}} = \frac{5 \times \$34{,}174 - \$6{,}300 - 5 \times \$3{,}826}{\$6{,}300 + 5 \times \$3{,}826} \approx +\mathbf{500}\%.$$

False-Alarm Cost Analysis

At a false-positive rate of 3% for hard-hat detection at distances >10 m (sub-60-pixel head height), and assuming 200 worker–camera observation events per day (single operational shift), approximately 6 false alarms per day require supervisor verification. At 5 min review time per event and \$30/h effective labour cost: $6 \times \frac{5}{60} \times \$30 \times 365 = \$5{,}475/\mathrm{yr}$ in false-alarm overhead.

Fine-tuning the detection head on 2,000 site-specific annotated images reduces the false-positive rate from 3% to ≤1% and costs approximately \$800 (one-time: annotation effort + GPU compute), saving \$3,650/yr in supervisor review time. This fine-tuning investment pays back within 90 days and is recommended as a standard post-deployment step for all safety-critical PPE monitoring systems.

Sensitivity Analysis

Break-even timing sensitivity to key cost parameters:

Parameter	Elasticity	Implication
Electricity cost (per kWh)	0.04	Insensitive; doubles cost shifts T^* by <0.1 months
Avoided incidents per year	1.00	Highly sensitive; see threshold below
Software licence cost	0.22	Moderately sensitive

Deployments with a historically low incident rate (<0.5 avoided incidents/yr) do not justify edge CapEx commitment; a cloud-hosted analytics service with per-alert billing is preferable for such sites, avoiding upfront hardware investment while still delivering PPE monitoring capability.

15 Foundation Model Adaptation: Technical Details and Edge Feasibility Pathway

Section 12.1 of the main article outlines the foundation model adaptation challenge and three feasibility directions. This section provides the complete technical treatment of the current hardware gap, the knowledge distillation objective, parameter-efficient fine-tuning, and sequential task implications.

15.1 Current Hardware Gap

Large vision–language models (CLIP ViT-B/32, BLIP-2) and segment-anything foundation models (SAM-2, EdgeTAM) provide generalist zero-shot capabilities—querying a scene with arbitrary natural-language prompts rather than a fixed class list—but require hardware that exceeds most deployed edge platforms:

- **CLIP ViT-B/32**: 151 M parameters; ≈4 GB VRAM required for real-time FP16 operation.
- **SAM-2**: image encoder requires ≥8 GB VRAM; precludes Jetson Orin NX 16 GB under concurrent multi-stream load.
- **YOLO-World** (open-vocabulary, decoupled image–text encoder): achieves 35 FPS on RTX 4090 (FP16) but falls to <5 FPS on Jetson Orin NX, illustrating the remaining edge gap even for architecturally distilled open-vocabulary detectors.

These requirements exceed all platforms in Table 10 of the main article except the Jetson AGX Orin 64 GB. The TSS survey (Zhou et al., ACM CSUR, vol. 58(3), 2025) identifies zero-shot anomaly detection as the highest-priority application for foundation models in traffic surveillance, and explicitly notes the same hardware gap as the primary near-term barrier.

15.2 Knowledge Distillation for Edge Adaptation

Knowledge distillation (KD) transfers the generalisation capability of a large foundation model (teacher T) into a compact, edge-deployable student architecture (S). The standard Hinton KD objective is:

$$\mathcal{L}_{\mathrm{KD}} = (1-\alpha)\,\mathcal{L}_{\mathrm{CE}}(S, y) \;+\; \alpha\,\tau^2\,\mathrm{KL}\Big(\sigma\big(\tfrac{z_T}{\tau}\big) \;\Big\|\; \sigma\big(\tfrac{z_S}{\tau}\big)\Big)\,, \tag{6}$$

where z_T, z_S are teacher and student logits; $\tau > 1$ is the softening temperature; $\alpha \in (0, 1)$ balances the hard-label cross-entropy $\mathcal{L}_{\mathrm{CE}}$ and the soft-label KL divergence; and $\sigma(\cdot)$ is the softmax function.

A higher temperature τ reveals more of the teacher's *dark knowledge*—the inter-class similarity structure encoded in near-zero logit entries that carry information beyond the hard label. When distilling from a foundation-model zero-shot classifier (whose teacher distribution captures rich inter-class structure learned from internet-scale data) to a compact domain-specific edge detector

(with limited training data and sparse inter-class structure), empirical results support $\tau \in [3, 8]$ and $\alpha \in [0.5, 0.9]$ for YOLO-family student architectures.

An important practical complication is that generative augmentation of the KD training set using diffusion models—to cover rare target classes or adverse visual conditions—introduces characteristic failure modes when the generated images exhibit distribution shift from the deployment domain (Bae et al., 2023). Practitioners should validate diffusion-synthesised training samples against real production-camera frames before integrating them into the KD pipeline, as spurious correlations in synthetic images can degrade student generalisation on real data.

15.3 Parameter-Efficient Fine-Tuning

Parameter-efficient fine-tuning (PEFT) methods—Low-Rank Adaptation (LoRA), prefix tuning, and adapter layers—enable site-specific domain adaptation of foundation models using <1% of the full-model parameter count. This makes periodic fine-tuning feasible as a cloud-tier batch job (overnight GPU compute), without requiring raw video data to be uploaded from edge devices. The resulting adapted parameters are pushed to edge nodes as lightweight delta files (typically <50 MB for LoRA rank-8 adapters on YOLO-scale architectures), maintaining the edge-only data architecture while enabling continuous model improvement from production deployment experience.

15.4 Hardware–Software Co-Design Trajectory

Next-generation edge accelerators add explicit hardware support for the operators that currently bottleneck foundation model deployment:

- **Hailo-10H** (26 TOPS) extends attention operator coverage beyond Hailo-8's CNN-centric kernel library.
- **Jetson AGX Orin DLA v4.0** adds native support for multi-head self-attention and dynamic input shapes previously requiring GPU fallback.
- **INT4 weight quantisation** (supported in TRT-LLM and MLC-LLM) halves the memory footprint of VLM language decoder components versus INT8, enabling Florence-2-class models on AGX Orin.

At current semiconductor roadmap pace, the hardware gap between foundation model requirements and Jetson-class device capacity is projected to close within approximately two accelerator generations (2027–2028).

15.5 Sequential and Temporal Task Implications

The domain-adaptation challenge is further compounded for *sequential* video understanding tasks. Deep learning pipelines for variable-length gesture and sign language recognition (Toshpulatov et al., 2025) demonstrate that temporal modelling across variable-length sequences faces the same domain-shift brittleness observed in standard frame-level video analytics: models trained on laboratory-collected datasets lose 15–30% accuracy when deployed on production camera feeds without site-specific fine-tuning.

Architectural solutions developed for this problem—attention-based temporal encoders and augmentation over temporally cropped sub-sequences—transfer directly to activity recognition and action classification in retail, industrial, and security verticals. This suggests that the sign-language recognition research community and the edge video analytics community share a common adaptation challenge whose joint treatment (shared benchmark datasets, standardised temporal augmentation protocols) would accelerate progress in both fields.

All scripts, model weights, environment specifications, and raw result CSVs referenced in this Supplementary Material are available at:

Repository: https://github.com/muxiddin19/EdgeVA